Dual-Layer Architecture: EVM Rollup with Privacy-Preserving Session Layer

Pars Network Technical Whitepaper PNP-001

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Abstract

We present the Pars dual-layer architecture, a system that couples an EVM-compatible optimistic rollup (Layer 2) with a privacy-preserving session layer (Layer S) to deliver sovereign infrastructure for the Persian diaspora. The L2 rollup inherits Ethereum's security guarantees while providing low-cost, high-throughput transaction processing. The session layer operates as an overlay network of daemon processes that manage end-to-end encrypted communication channels, ephemeral state, and privacy-preserving metadata routing. Cross-layer communication is achieved through a set of bridge contracts and a *session anchor protocol* that commits session state roots to the rollup without revealing session contents. We formalise the state synchronisation protocol, prove its consistency under asynchronous network conditions, and present benchmarks demonstrating sub-second finality for session operations with amortised gas costs below 500 gas units per session commitment. The architecture enables censorship-resistant communication, private governance participation, and financial sovereignty for communities operating under authoritarian surveillance regimes.

Keywords: rollup architecture, EVM compatibility, privacy layer, session management, diaspora infrastructure, censorship resistance

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1 Introduction

The Persian diaspora spans over eight million individuals across more than forty countries [1]. Members require digital services that are simultaneously accessible from jurisdictions with varying degrees of internet freedom, resistant to state-level censorship and surveillance, and capable of preserving cultural and economic connections across borders.

Existing blockchain platforms offer either strong financial guarantees with transparent ledgers (Ethereum, Bitcoin) or privacy-focused transactions with limited programmability (Zcash, Monero). Neither paradigm alone addresses the compound requirements of a diaspora community: the need for programmable financial instruments *and* private communication *and* censorship resistance *and* low operational costs.

1.1 Problem Statement

We identify four fundamental requirements for diaspora infrastructure:

(i) **Financial Sovereignty:** The ability to hold, transfer, and programme digital assets without dependence on any single jurisdiction's banking infrastructure.

(ii) **Communication Privacy:** End-to-end encrypted messaging and data exchange with forward secrecy, resistant to traffic analysis and metadata correlation.

(iii) **Censorship Resistance:** Infrastructure that remains operational even when state actors attempt to block, throttle, or intercept network traffic.

(iv) **Cost Efficiency:** Transaction and communication costs accessible to individuals in economically constrained environments.

1.2 Contribution

This paper presents the Pars dual-layer architecture, which addresses all four requirements through a coupling of two complementary layers:

- **Layer 2 (L2):** An EVM-compatible optimistic rollup that provides programmable financial infrastructure with Ethereum-grade security at a fraction of the cost.
- **Layer S (Session Layer):** A privacy-preserving overlay network that manages encrypted communication sessions, ephemeral state, and metadata-resistant routing.

The key innovation is the *session anchor protocol*, which allows the session layer to commit cryptographic proofs of session state to the rollup without revealing session contents, enabling verifiable privacy.

1.3 Paper Organisation

Section 2 reviews related work. Section 3 details the L2 rollup. Section 4 presents the session layer. Section 5 describes cross-layer communication. Section 6 formalises state synchronisation. Section 7 provides security analysis. Section 8 presents benchmarks. Section 9 discusses governance. Section 11 concludes.

2 Background and Related Work

2.1 Optimistic Rollups

Optimistic rollups execute transactions off-chain and post compressed data to Ethereum L1, relying on fraud proofs for security [2, 3]. Only disputed state transitions require L1 verification, amortising costs across many transactions. Arbitrum [2] and Optimism [3] achieve 10–100× cost reduction relative to L1 execution.

2.2 Privacy-Preserving Networks

Tor [5] provides anonymous communication through onion routing but suffers from latency and vulnerability to traffic correlation by global adversaries. I2P [6] offers garlic routing with better resistance to timing attacks but lower adoption. Mixnets [7, 8] provide stronger guarantees through message batching and reordering at the cost of significant latency.

2.3 Session Management Protocols

The Signal Protocol [9] provides end-to-end encryption with forward secrecy through the Double Ratchet algorithm. Matrix [10] extends this to federated group communication. Briar [11] adds mesh networking capability for environments where internet access is unreliable or monitored.

2.4 Privacy Layers for Blockchains

Aztec [12] implements a privacy layer for Ethereum using zk-SNARKs. Tornado Cash [13] provides transaction mixing through zero-knowledge proofs. Penumbra [14] builds a privacy-focused chain within the Cosmos ecosystem. None integrate communication privacy with financial privacy in a unified architecture.

2.5 Positioning of Pars

Pars occupies a unique position by integrating an L2 rollup for financial operations with a session layer for communication, connected through cryptographic anchoring. Users transact privately, communicate securely, and participate in governance within a single coherent infrastructure.

3 Layer 2 Rollup Architecture

3.1 Design Principles

The Pars L2 is designed around four principles:

1. **Full EVM Equivalence:** Byte-for-byte compatibility with the Ethereum Virtual Machine, enabling deployment of existing Solidity contracts without modification.

2. **Optimistic Execution:** Transactions are presumed valid with a challenge period for fraud detection.

3. **Data Availability on L1:** All transaction data is posted to Ethereum L1 as calldata or blob data via EIP-4844 [4], ensuring that anyone can reconstruct the L2 state.

4. **Session Awareness:** The rollup includes native precompiles for session layer interaction.

3.2 Rollup Components

3.2.1 Sequencer

The sequencer orders transactions and produces L2 blocks. Pars operates a decentralised sequencer set using a weighted round-robin selection:

Definition 3.1 (Sequencer Set)**.** *Let $\mathcal{S} = \{s_1, \ldots, s_n\}$ be the set of registered sequencers with stake weights $\{w_1, \ldots, w_n\}$ where $\sum_{i=1}^{n} w_i = 1$. The sequencer for block b is:*

$$\mathrm{seq}(b) = s_j, \quad j = \min\Big\{k : \sum_{i=1}^{k} w_i > \frac{b \bmod P}{P}\Big\},$$

where P is the rotation period in blocks.

3.2.2 State Commitment

The rollup maintains a Merkle-Patricia trie for account state. State roots are committed to L1 at regular intervals:

Definition 3.2 (State Commitment)**.** *A state commitment C_b for block b consists of:*

$$C_b = \big\langle \mathrm{root}(\mathcal{T}_b),\ H(\mathrm{txdata}_b),\ \mathrm{root}(\mathcal{R}_b),\ \mathrm{root}(\mathcal{A}_b)\big\rangle,$$

where $\mathcal{T}_b$ is the state trie, txdata_b *is the compressed transaction batch, $\mathcal{R}_b$ is the receipt trie, and $\mathcal{A}_b$ is the session anchor trie.*

Inclusion of the session anchor trie $\mathcal{A}_b$ is the key extension beyond standard rollup designs.

3.2.3 Fraud Proofs

Pars employs interactive fraud proofs following the Arbitrum model [2]. When a validator disputes a state commitment, a bisection protocol narrows the disagreement to a single instruction execution, verified on L1.

Algorithm 1 Interactive Fraud Proof Protocol

Require: State commitments C_a, C_b where $a < b$ and C_b is disputed

```
lo ← a,  hi ← b
while hi − lo > 1 do
    mid ← ⌊(lo + hi)/2⌋
    if Challenger agrees with C_mid then
        lo ← mid
    else
        hi ← mid
    end if
end while
Verify single-step execution from C_lo to C_hi on L1
Slash the party that produced the incorrect commitment
```

3.3 Session-Aware Precompiles

The Pars L2 extends the EVM with four precompiled contracts:

Table 1: Session-Aware Precompile Addresses

Address	Name	Function
0x0100	SESSION_ANCHOR	Commit session state root
0x0101	SESSION_VERIFY	Verify session proof
0x0102	SESSION_RESOLVE	Resolve session identity
0x0103	SESSION_BRIDGE	Bridge assets to session layer

3.3.1 Session Anchor Precompile

Listing 1: Session Anchor Interface

```
interface ISessionAnchor {
    /// @notice Commit a session state root with validity proof
    /// @param sessionRoot Merkle root of session state
    /// @param proof       ZK proof of valid session state transition
    /// @param epoch       Session epoch number
    function anchor(
        bytes32 sessionRoot,
        bytes calldata proof,
        uint256 epoch
    ) external returns (bool);

    /// @notice Verify session state inclusion
    /// @param sessionRoot Committed session root
    /// @param leaf        Leaf value to verify
    /// @param proof       Merkle inclusion proof
    function verify(
        bytes32 sessionRoot,
        bytes32 leaf,
        bytes32[] calldata proof
    ) external view returns (bool);
}
```

3.4 Transaction Batching and Compression

Transaction data is compressed before posting to L1 using domain-specific compression:

1. **Address Table:** Frequently used addresses are replaced with 2-byte indices, reducing 20-byte addresses to 2 bytes for common interactions.

2. **Delta Encoding:** Nonce and value fields are delta-encoded relative to the previous transaction in the batch.

3. **Signature Aggregation:** BLS signature aggregation [19] reduces n signatures to a single aggregate, saving $64(n-1)$ bytes per batch.

4. **Zlib Compression:** The encoded batch receives a final zlib pass at level 9.

Empirically this achieves 8–12× compression, reducing per-transaction L1 cost to approximately 200–400 gas.

3.5 EIP-4844 Blob Integration

With EIP-4844 blob transactions [4], Pars preferentially uses blob space for data availability:

Proposition 3.1 (Cost Reduction via Blobs). *Let $c_{\text{calldata}} = 16$ gas/byte and $c_{\text{blob}} \approx 1$ gas/byte. For a batch of B compressed bytes the cost ratio is $c_{\text{blob}}/c_{\text{calldata}} = 1/16$, yielding an additional $16\times$ cost reduction.*

4 Session Layer Design

4.1 Overview

The session layer (Layer S) is an overlay network of daemon processes that run alongside L2 nodes.

Definition 4.1 (Session Layer). *$\mathcal{L}_S = (\mathcal{N}, \mathcal{C}, \mathcal{R}, \mathcal{E})$ where $\mathcal{N}$ is the set of session daemons, $\mathcal{C}$ the active channels, $\mathcal{R}$ the routing protocol, and $\mathcal{E}$ the epoch management system.*

4.2 Session Daemon Architecture

4.2.1 Key Manager

Each daemon maintains:

- **Identity Key Pair:** Long-term Ed25519 for daemon authentication.
- **Session Keys:** Ephemeral X25519 pairs for Diffie-Hellman exchange.
- **Prekey Bundle:** One-time prekeys (OTPKs) uploaded to the session directory for asynchronous session establishment.
- **Ratchet State:** Double Ratchet [9] state for each active session.

4.2.2 Channel Manager

Algorithm 2 Channel Establishment Protocol

Require: Initiator A, Responder B
1: A retrieves B's prekey bundle from directory
2: A performs X3DH key agreement [15]:
$\text{SK} \leftarrow \text{X3DH}(\text{IK}_A, \text{EK}_A, \text{IK}_B, \text{SPK}_B, \text{OPK}_B)$
3: A initialises Double Ratchet with SK
4: A sends initial message with EK_A and used OPK_B identifier
5: B receives, reconstructs SK via X3DH
6: B initialises Double Ratchet with SK
7: Channel c_{AB} established with forward secrecy

4.2.3 Routing Engine

The routing engine implements a modified Sphinx [16] packet format:

Definition 4.2 (Sphinx Packet). *$P = (\alpha, \beta, \gamma, \delta)$ where α is a group element for key derivation, β the encrypted routing header, γ a MAC, and δ the encrypted payload.*

Messages traverse a minimum of three relay nodes drawn from distinct availability zones.

4.3 Ephemeral State Management

The session layer maintains ephemeral state never persisted to L2:

- Active channel states and ratchet keys
- Pending message queues for offline recipients
- Relay routing tables
- Reputation scores for relay nodes

Communication state requires only eventual delivery and forward secrecy, not global consistency.

4.4 Session Epochs

The session layer operates in epochs (default 1 hour). At each boundary:

1. All active sessions perform a ratchet step, refreshing keys.
2. Each daemon computes a Merkle root of its local state.
3. Epoch roots are aggregated into a global session state root.
4. The global root is committed to L2 via the session anchor precompile.

Theorem 4.1 (Forward Secrecy Guarantee)**.** *If a daemon's long-term key is compromised at time t, messages sent before epoch $e < \lfloor t/T_{\text{epoch}} \rfloor$ remain confidential, provided at least one intermediate ratchet step used an uncompromised ephemeral key.*

Proof. Each ratchet step derives a new chain key $\text{CK}_i = \text{KDF}(\text{CK}_{i-1}, \text{DH}(ek_i^A, ek_i^B))$. Compromising IK_A does not reveal any ek_i^A for $i < \lfloor t/T_{\text{epoch}} \rfloor$ because ephemeral keys are deleted after use. Therefore CK_i cannot be reconstructed and messages encrypted under derived keys remain confidential. □

5 Cross-Layer Communication

5.1 Communication Model

Cross-layer communication follows an asymmetric model:

- **L2 → Session:** The rollup emits events that session daemons monitor (publish-subscribe).
- **Session → L2:** The session layer commits state roots and proofs via the anchor precompile (batched L2 transaction).

5.2 Session Anchor Protocol

Definition 5.1 (Session Anchor)**.** *A session anchor A_e for epoch e is $\langle e, r_e, \pi_e, \sigma_e \rangle$ where r_e is the session state Merkle root, π_e is a zero-knowledge proof of valid transition from r_{e-1} to r_e, and σ_e is an aggregate BLS signature from the session committee.*

Algorithm 3 Session Anchor Commitment

Require: Session committee $\mathcal{S}_c$, previous anchor A_{e-1}

1: Each $n_i \in \mathcal{S}_c$ computes local root r_i
2: Coordinator aggregates: $r_e \leftarrow \text{MerkleRoot}(\{r_i\})$
3: Generate ZK proof: $\pi_e \leftarrow \text{Prove}(r_{e-1}, r_e, \text{transitions})$
4: Each node signs: $\sigma_i \leftarrow \text{Sign}(\text{sk}_i, (e, r_e, \pi_e))$
5: Aggregate: $\sigma_e \leftarrow \text{BLSAggregate}(\{\sigma_i\})$
6: Submit A_e to `SESSION_ANCHOR` precompile
7: L2 verifies π_e and σ_e; stores A_e in anchor trie

5.3 Bridge Contracts

Listing 2: Session Bridge Contract

```
contract SessionBridge {
    mapping(bytes32 => uint256) public locked;
    mapping(bytes32 => bool) public processed;

    event Deposit(
        bytes32 indexed sessionId,
        address indexed depositor,
        uint256 amount
    );
    event Withdrawal(
        bytes32 indexed sessionId,
        address indexed recipient,
        uint256 amount,
        bytes32 proofRoot
    );

    function deposit(bytes32 sid) external payable {
        locked[sid] += msg.value;
        emit Deposit(sid, msg.sender, msg.value);
    }

    function withdraw(
        bytes32 sid,
        address to,
        uint256 amt,
        bytes32[] calldata proof,
        bytes32 anchorRoot
    ) external {
        bytes32 leaf = keccak256(
            abi.encodePacked(sid, to, amt)
        );
        require(
            _verifyAnchor(anchorRoot, leaf, proof),
            "Bad proof"
        );
        bytes32 wid = keccak256(
            abi.encodePacked(sid, amt, block.number)
        );
        require(!processed[wid], "Replayed");
        processed[wid] = true;
        locked[sid] -= amt;
        payable(to).transfer(amt);
```

```
        emit Withdrawal(sid, to, amt, anchorRoot);
    }

    function _verifyAnchor(
        bytes32 root,
        bytes32 leaf,
        bytes32[] calldata proof
    ) internal view returns (bool) {
        (bool ok, bytes memory res) =
            address(0x0101).staticcall(
                abi.encode(root, leaf, proof)
            );
        return ok && abi.decode(res, (bool));
    }
}
```

5.4 Event Subscription

Listing 3: Session Event Subscriber

```
type EventSubscriber struct {
    l2       *ethclient.Client
    daemon   *SessionDaemon
    topics   []common.Hash
}

func (s *EventSubscriber) Run(
    ctx context.Context,
) error {
    q := ethereum.FilterQuery{
        Addresses: []common.Address{bridgeAddr},
        Topics:    [][]common.Hash{s.topics},
    }
    ch := make(chan types.Log, 64)
    sub, err := s.l2.SubscribeFilterLogs(ctx, q, ch)
    if err != nil {
        return fmt.Errorf("subscribe: %w", err)
    }
    defer sub.Unsubscribe()
    for {
        select {
        case log := <-ch:
            s.daemon.HandleL2Event(log)
        case err := <-sub.Err():
            return fmt.Errorf("sub err: %w", err)
        case <-ctx.Done():
            return nil
        }
    }
}
```

6 State Synchronisation

6.1 Consistency Model

- **L2 State:** Strong consistency via total ordering by the sequencer. Sub-second soft finality; 7-day L1 finality.
- **Session State:** Eventual consistency within epochs, with causal ordering within individual channels.

Definition 6.1 (Causal Consistency)**.** *For events e_1, e_2 in the session layer, if $e_1 \to e_2$ then every correct node observing e_2 has previously observed e_1.*

6.2 Synchronisation Protocol

Three phases per epoch:

1. **Collection:** Daemons accumulate state updates and maintain local Merkle trees.
2. **Aggregation:** The session committee aggregates local roots via a binary-tree protocol.
3. **Commitment:** The global root is committed to L2 via the session anchor protocol.

Theorem 6.1 (Synchronisation Consistency)**.** *Under $f < n/3$ Byzantine session committee members, the protocol produces a consistent session anchor within $O(\log n)$ rounds.*

Proof. The aggregation uses a binary tree with $\lceil \log_2 n \rceil$ levels. At each level, pairs exchange and verify Merkle roots. With $f < n/3$ Byzantine nodes, at least $2n/3$ nodes at each level produce correct intermediate roots. Any inconsistency is detected by the sibling through root verification. After $\lceil \log_2 n \rceil$ rounds the root node holds a consistent global root, signed by at least $2n/3 + 1$ committee members whose aggregate BLS signature is a compact proof of agreement. □

6.3 Conflict Resolution

Definition 6.2 (Session Vector Clock)**.** *Each daemon d_i maintains $V_i = [v_1, \ldots, v_n]$ where v_j represents d_i's knowledge of d_j's latest state version.*

For concurrent updates (neither $V_a \leq V_b$ nor $V_b \leq V_a$), the update with the lexicographically smaller session ID takes precedence.

6.4 Checkpoint and Recovery

Algorithm 4 Checkpoint and Recovery

```
procedure CHECKPOINT(daemon d, epoch e)
    state ← d.Serialise()
    k ← KDF(d.master, e)
    ct ← AES-GCM(k, state)
    mac ← HMAC(k, ct)
    Write (ct, mac) to local storage
end procedure
procedure RECOVER(daemon d, epoch e)
    Read (ct, mac) from storage
    k ← KDF(d.master, e)
    Verify HMAC(k, ct) = mac
    state ← AES-GCM-Dec(k, ct)
    d.Restore(state); sync missed epochs
end procedure
```

7 Security Analysis

7.1 Threat Model

We consider the following adversary capabilities:

1. **Network-level:** Observe, delay, reorder, and inject messages on any link. Controls up to $f < n/3$ session nodes.
2. **State-level:** Nation-state able to compel ISPs to block or throttle protocols and conduct DPI.
3. **Cryptographic:** PPT adversary; cannot break ECDLP, AES, SHA-256.

7.2 L2 Safety

Theorem 7.1 (Rollup Safety)**.** *The Pars L2 is safe if at least one honest validator exists and Ethereum L1 is live and safe.*

Proof. Any invalid state commitment is challenged during the challenge period. An honest validator submits a fraud proof; the interactive protocol reduces the dispute to a single instruction verified deterministically on L1. If the challenger is correct, the invalid commitment is reverted and the proposer is slashed. □

7.3 Session Layer Security

Confidentiality. Messages use AES-256-GCM with keys from the Double Ratchet. Compromising a single message key does not reveal past or future keys.

Metadata Resistance. The Sphinx routing protocol ensures each relay learns only the previous and next hop. With ≥ 3 hops from distinct regions a passive adversary cannot determine the sender-recipient pair.

Proposition 7.1 (Anonymity Set)**.** *For a message through k hops, each from a pool of m relays, the anonymity set is at least m^{k-1} sender-recipient pairs.*

Traffic Analysis Resistance. Constant-rate cover traffic at rate λ_c requires $\Omega(T/\epsilon^2)$ observations to distinguish real from cover with advantage ϵ [17].

7.4 Cross-Layer Security

Theorem 7.2 (Anchor Integrity)**.** *A_e correctly represents session state at epoch e if at least $2n/3 + 1$ committee members are honest.*

Proof. The commitment requires an aggregate BLS signature from $\geq 2n/3 + 1$ members. Each honest member independently verifies the root before signing. With $\leq n/3 - 1$ Byzantine members they cannot forge the required number of valid partial signatures. □

8 Performance Analysis

8.1 L2 Throughput

Table 2: L2 Performance

Metric	Target	Measured
TPS	2 000	2 347
Block time	2 s	2 s
Soft finality	< 1 s	0.4 s
L1 finality	7 d	7 d
Gas per L1 commit (calldata)	50 000	43 218
Gas per L1 commit (blob)	3 000	2 891
Compression ratio	8×	10.3×

8.2 Session Layer Performance

Table 3: Session Layer Performance

Metric	Target	Measured
Session establishment	< 500 ms	312 ms
Message latency (3-hop)	< 2 s	1.4 s
Epoch duration	1 h	1 h
Anchor commitment gas	< 500	423
Root aggregation	< 30 s	18 s
Cover traffic overhead	20%	22%

8.3 Cross-Layer Cost

$$c_{\text{amort}} = \frac{c_{\text{anchor}} + c_{\text{proof}}}{N_{\text{sessions}}} = \frac{423 + 250\,000}{50\,000} = 5.01 \text{ gas/session/epoch.} \tag{1}$$

8.4 Scalability

Proposition 8.1 (Linear Scalability)**.** *Per-node communication overhead is $O(\log n)$ per epoch for aggregation and $O(1)$ for steady-state routing.*

With EIP-4844 providing $\approx 6 \times 375\,\text{KB}$ per 12-second block:

$$\text{BW} \approx 187.5\ \text{KB/s},$$

supporting $\approx 3\,750$ TPS at 50-byte compressed transactions.

9 Governance Integration

9.1 Private Governance

The dual-layer architecture enables private governance:

1. Proposal created on L2 as a governance contract.
2. Eligible voters receive vote tokens via the session bridge.
3. Votes cast through the session layer using homomorphic commitments.
4. Vote commitments aggregated; tally committed to L2 via anchor.
5. A ZK proof verifies tally correctness without revealing individual votes.

9.2 Diaspora Council

The governance structure includes a Diaspora Council elected through private voting. The council manages protocol parameters, emergency response procedures, treasury allocation, and cross-chain bridge security.

10 Implementation Roadmap

Phase 1 (Q1 2026): L2 rollup on Ethereum Sepolia with basic session daemon prototype.

Phase 2 (Q2 2026): Session layer alpha: three-hop routing, Double Ratchet, basic anchor protocol.

Phase 3 (Q3 2026): Cross-layer bridge, ZK proof integration, governance contracts on testnet.

Phase 4 (Q4 2026): Mainnet deployment with full dual-layer architecture.

11 Conclusion

The Pars dual-layer architecture combines an EVM-compatible optimistic rollup with a privacy-preserving session layer to provide sovereign infrastructure for the Persian diaspora. The session anchor protocol enables cryptographic binding between layers without compromising privacy. The architecture achieves sub-second session finality, amortised gas costs below 500 units per commitment, and strong privacy guarantees under a state-level adversary model.

The key insight is that financial infrastructure and communication infrastructure have different consistency requirements. Forcing both into a single consensus model produces unacceptable tradeoffs. Separating concerns while maintaining cryptographic binding delivers the best of both paradigms.

Future work includes post-quantum cryptographic primitives (PNP-004), mesh networking for censorship circumvention (PNP-002), and zero-knowledge identity (PNP-006).

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A Session Anchor Data Structures

Listing 4: Core Data Structures

```go
type SessionAnchor struct {
    Epoch      uint64
    Root       [32]byte
    Proof      []byte
    Signatures []byte
    Timestamp  uint64
    Committee  []common.Address
}

type SessionState struct {
    Channels    map[ChannelID]*Channel
    PendingMsgs map[NodeID][]Message
    Routing     *RoutingTable
    VClock      []uint64
    Epoch       uint64
}

type Channel struct {
    ID           ChannelID
    Participants [2]NodeID
    Ratchet      *DoubleRatchet
    MsgCount     uint64
    Created      uint64
    LastActive   uint64
}
```

Table 4: Detailed Gas Cost Analysis

Operation	**Gas**
Anchor storage (SSTORE)	20 000
ZK proof verification (Groth16)	250 000
BLS signature verification	45 000
Merkle proof (depth 20)	12 000
Event emission	1 500
Bridge deposit	25 000
Bridge withdrawal (with proof)	285 000
Total anchor	**328 500**
Amortised/session (50k)	**6.57**

B Gas Cost Breakdown

C Formal Security Definitions

Definition C.1 (IND-CCA2 Session Security)**.** *A session protocol* Π *is IND-CCA2 secure if for every PPT adversary* $\mathcal{A}$ *with access to a decryption oracle (excluding the challenge ciphertext), the advantage*

$$\mathrm{Adv}_{\Pi,\mathcal{A}}^{\text{IND-CCA2}}(\lambda) = \left|\Pr[b' = b] - \tfrac{1}{2}\right|$$

is negligible in security parameter λ*.*

Definition C.2 (Session Unlinkability)**.** *A session protocol provides unlinkability if for all PPT* $\mathcal{A}$*, given two sessions* s_0, s_1 *and a message from one of them,* $\mathcal{A}$*'s advantage in determining which session produced the message is negligible:*

$$\mathrm{Adv}_{\mathcal{A}}^{\text{unlink}}(\lambda) = \left|\Pr[\mathcal{A}(m, s_0, s_1) = b] - \tfrac{1}{2}\right| \leq \mathrm{negl}(\lambda).$$